

Shalini Majumder

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EDUCATION

- 2025 - Present MSc in Bioinformatics, **University of Birmingham, UK**
Thesis: *PCS-Guided Multi-Omics Classification of Prostate Cancer Using Explainable Machine Learning and Network-Based Drug Repurposing*, under supervision of Dr Mamunur Rashid, University of Birmingham, UK
- 2023 - 2025 MSc in Genetics (Gold Medalist, First Class First), A+ Grade, **University of Calcutta** (CGPA: 8.647/10.0)
Thesis: *In silico Characterization and Tissue-specific Expression Analysis of Mouse lncRNA Gm26809*, under supervision of Dr Arpita Konar, Presidency University, Kolkata
- 2020 - 2023 BSc in Zoology (Honours), A++ Grade, **Bethune College, University of Calcutta** (CGPA: 9.041/10.0)

RESEARCH EXPERIENCE

MSc Thesis Project

(Feb 2026 – present)

- Currently working under the supervision of **Dr Mamunur Rashid**, Assistant Professor, Department of Cancer and Genomic Sciences, University of Birmingham, UK
- Research Topic:** Multi-Omic Integration and Epigenetic Profiling of Prostate Cancer Subtypes. The project investigates prostate cancer heterogeneity by integrating RNA-seq and DNA methylation data from TCGA and PCTA cohorts. It focuses on PCS subtype classification using machine learning (Random Forest, centroid-based methods) and explainable AI (SHAP) to identify biologically relevant biomarkers. The study further explores methylation–expression regulatory relationships using mediation analysis, alongside pathway enrichment, network analysis, and drug repurposing, implemented through reproducible computational workflows in Python and R.

Group Research Project

(2025 - present)

- Currently working under the supervision of **Dr Lindsey Compton**, Associate Professor in Genetics, School of Biosciences, University of Birmingham, UK
- Research Topic:** Optimising GWAS pipelines using R to uncover genetic loci underlying canopy architecture traits and yield (complex traits) in tetraploid potato using SNP and phenotyping datasets, enabling enhanced precision breeding for climate-resilient crop improvement. The work integrates population structure analysis (DAPC), environmental correction using Best Linear Unbiased Estimates (BLUES), and comparative statistical models to improve trait–locus associations.

Research Internship

(2023 - 2025)

- Pursued under the supervision of **Dr Tapan Chowdhury**, Head and Associate Professor, Department of Computer Science and Engineering, Techno Main Salt Lake, Kolkata, India
- Research Domain:** Signaling Pathway Analysis, Molecular Docking, Deep Learning for Protein-Ligand Binding Affinity Prediction

Summer Research Internship

(2024)

- Under the supervision of **Dr Arpita Konar**, Assistant Professor, Institute of Health Sciences, Presidency University, Kolkata
- Dissertation Topic:** *In silico* Characterization and Tissue-specific Expression Analysis of Mouse lncRNA Gm26809. Used various bioinformatics tools (e.g., RNAfold, CPC2, lncLocator, lncRRISearch, etc.) to characterize the lncRNA and used qPCR to quantify and validate the tissue-specific expression profiling.

TEACHING & MENTORING EXPERIENCE

Undergraduate Research Project Co-Supervisor

(Aug 2025 - Present)

Techno Main Salt Lake, India

- Co-supervising a group of B.Tech final-year students under the supervision of **Dr Tapan Chowdhury**, Head and Associate Professor, Department of Computer Science and Engineering, Techno Main Salt Lake, Kolkata, India
- Conducting regular remote mentoring sessions from the UK to guide literature review, research design, and computational methodology
- Guided students in conducting a **miRNA–mRNA interaction network analysis for Alzheimer’s disease biomarker discovery**
- Mentored students in scientific writing and manuscript preparation; the work resulted in a **student-led bioRxiv preprint currently under review at *Journal of Molecular Neuroscience***

PUBLICATIONS-JOURNAL

- [1] **S. Majumder**, E. Lodh, T. De, and T. Chowdhury, “In silico Characterization of Melanoma-derived Mouse lncRNA Gm26982 and Comparative Analyses with Human Ortholog,” *Molecular Genetics and Genomics*, vol. 301, no. 103, p. 25, Apr. 2026, doi: [10.1007/s00438-026-02423-1](https://doi.org/10.1007/s00438-026-02423-1).
- [2] E. Lodh, T. Chowdhury, and **S. Majumder**, “BioAlignNet: A GPU-Accelerated Framework for Efficient Global Sequence Alignment Across Genomics and Proteomics Scales,” *Computational Intelligence*, vol. 42, no. 3, p. e70232, Apr. 2026, doi: [10.1111/coin.70232](https://doi.org/10.1111/coin.70232).

- [3] E. Lodh, T. Chowdhury, A. Shaw, D. Bedajna, S. Bera, **S. Majumder**, and M. De “A Stacked Ensemble Meta-Classifer Framework for PAM50 Breast Cancer Subtype Prediction from Gene Expression Signatures,” *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 15, no. 78, p. 31, Mar. 2026. doi: [10.1007/s13721-026-00755-x](https://doi.org/10.1007/s13721-026-00755-x).
- [4] E. Lodh, **S. Majumder**, T. Chowdhury, and M. De, “RLBindDeep: A ResNet-LSTM based novel framework for protein–ligand binding affinity prediction,” *Journal of Molecular Graphics and Modelling*, vol. 144, no. 109282, p. 24, Jan. 2026, doi: [10.1016/j.jmgm.2026.109282](https://doi.org/10.1016/j.jmgm.2026.109282)
- [5] **S. Majumder**, E. Lodh, and T. Chowdhury, “Implications of trinodal inhibitions and drug repurposing in MAPK pathway: A putative remedy for breast cancer,” *Computational Biology and Chemistry*, vol. 113, p. 108255, Oct. 2024, doi: [10.1016/j.compbiolchem.2024.108255](https://doi.org/10.1016/j.compbiolchem.2024.108255)
- [6] **E. Lodh**, S. Majumder, T. Chowdhury, and M. De, “NetPolicy-RL: Network-Informed Offline Reinforcement Learning for Pharmacogenomic Drug prioritization,” *Journal of Computer-Aided Molecular Design*, Apr. 2026, (Accepted, In Press).
- [7] T. Chowdhury, E. Lodh, and **S. Majumder**, “A Novel Method for Drug Application on Signaling Pathways with Concurrent Faults Using Probabilistic Boolean Networks,” *Computational Intelligence*, Feb. 2025, (Under Revision).
- [8] T. Chowdhury, S. Samanta, P. Sil, S. Banerjee, S. Banerjee, E. Lodh, **S. Majumder**, and M. De, “Biomarker Discovery For Breast Cancer Subtype Detection Using Deep Learning Based Approach,” *SN Computer Science*, Oct. 2025, (Under Review).
- [9] E. Lodh, T. Chowdhury, **S. Majumder**, M. De, and S. Singha, “SkinEnsemNet: A Novel Ensemble Deep Learning Framework for Skin Cancer Classification,” *Multimedia Tools and Applications*, Oct. 2025, (Under Review).
- [10] T. Chowdhury, A. Maitra, A. Agarwal, A. Sur, S. Sarkar, E. Lodh, and **S. Majumder**, “Multiple Fault Analysis and Drug Therapy on Signaling Pathways Using GPU-Accelerated Dynamic Bayesian Network-based Model,” *The Journal of Supercomputing*, Oct. 2025, (Submitted).
- [11] **S. Majumder**, M. Sengupta, “The Regulome in Disease: Gene Regulatory Mechanisms Beyond the Coding Genome”, *Mammalian Genome*, Feb. 2026 (Under Review).
- [12] E. Lodh, **S. Majumder**, and T. Chowdhury, “PAM50GenePath: A CatBoost-Driven Framework for Enhanced Invasive Breast Cancer Subtype Classification,” *Molecular Genetics and Genomics*, Apr. 2026, (Under Review).

PUBLICATION-CONFERENCE

- [1] E. Lodh, **S. Majumder** and T. Chowdhury, “CGDeepAff: Deep Learning-Based Approach for Protein-Ligand Binding Affinity Estimation Using CNN-GRU,” *2025 8th International Conference on Electronics, Materials Engineering & Nano-Technology (IEMENTech)*, Kolkata, India, 2025, pp. 1-6, doi: [10.1109/IEMENTech65115.2025.10959578](https://doi.org/10.1109/IEMENTech65115.2025.10959578). (**Presenter**)
- [2] E. Lodh, **S. Majumder**, T. De, T. Chowdhury, and M. De, “GeneDeepNet: A Differential Expression and Deep Learning Based Novel Framework for Detecting Invasive Breast Cancer Subtypes,” *4th International Conference on Machine Learning and Data Engineering (ICMLDE-2025)*, *Procedia Computer Science*, 2025 (Accepted, In Press).

PUBLICATIONS-PREPRINT

- [1] A. Ray, K. Agarwal, S. Jha, A. M. Singh, **S. Majumder**, E. Lodh, and T. Chowdhury, “miRNA–mRNA interaction network analysis in Alzheimer’s disease for biomarker discovery,” *bioRxiv (Cold Spring Harbor Laboratory)*, Jan. 2026. doi: [10.64898/2026.01.28.702349](https://doi.org/10.64898/2026.01.28.702349) (Manuscript under review at Journal of Molecular Neuroscience)

PUBLICATIONS-BOOK CHAPTER (INVITED)

- [1] S. Majumder, **E. Lodh**, and S. Datta, “Explainable AI in Redox Medicine: Challenges and Opportunities,” in *Oxidative Stress–Mediated Disorders: Advances in Diagnosis and Therapy with AI and Machine Learning*, edited by S. Roy, D. Ghosh, P. Kalita, and P. H. Guzzi, Springer Nature, 2026. (Submitted).

PROJECTS

MSc Coursework: Mathematical & Statistical Analysis using R ([GitHub](#)) (2026)

- Developed a comprehensive, reproducible analytical workflow in R to solve applied problems in probability, statistical inference, and mathematical modelling as part of MSc Bioinformatics coursework.
- Implemented modular and reusable functions (cw_functions.R) to streamline statistical computations and improve code efficiency and clarity.
- Performed hypothesis testing, distribution analysis, and numerical computations, with results interpreted through structured visualisations and statistical reasoning.
- Documented the complete workflow using R Markdown, ensuring full reproducibility and integration of code, results, and interpretation.
- Achieved a grade of 81.5%, demonstrating strong proficiency in translating theoretical concepts into computational solutions.

- EHR-Based Mortality and LOS Prediction using ML, Deep Learning, and Clinical LLMs (GitHub) (2026)
- Developed an end-to-end predictive modelling pipeline on synthetic Electronic Health Record (EHR) data inspired by MIMIC-III/IV, integrating patient demographics, admission details, and longitudinal diagnosis codes.
 - Performed advanced data preprocessing, including ICD-9 to ICD-10 harmonisation, temporal feature extraction (LOS/Length-of-stay), and clinically meaningful diagnosis grouping using CCSR categories.
 - Implemented and compared multiple modelling paradigms: traditional machine learning (XGBoost, Random Forest), sequential models (LSTM, Transformer), and clinical language models (BioClinicalBERT, Internist-7B).
 - Designed patient-level sequence representations to capture longitudinal clinical trajectories and prevent data leakage using grouped data splitting strategies.
 - Evaluated models using AUROC, F1-score, MAE, R^2 , and calibration metrics, alongside subgroup fairness analysis across demographic groups (e.g., race/ethnicity).
 - Demonstrated that well-engineered traditional ML models outperform more complex architectures for structured clinical prediction tasks, while LLMs exhibit strong discrimination but calibration limitations.

- DELLY_VCF_Trio Structural Variant (SV) Analysis Pipeline (GitHub) (2025)
- The pipeline processes a Delly-generated VCF, performs sample detection (child and parents), parent detection (who father and who mother), sex determination of the child, SV extraction, de novo detection, family-based inference, BED-based annotation, and high-quality visualisation.
 - It combines shell scripting, AWK and Python scripts, ANNOVAR and ClinVar databases, and BEDTools-based comparisons.
 - This can analyse a Delly VCF file of any family of trio (father, mother, child) for their structural variants. It can be applied to other Delly VCF files by tweaking the file and folder names a bit and changing the code.

RELEVANT EXPERIENCE

- AI Hackathon Volunteer, HealthTech AI Hub & University of Birmingham (2026)
- Supported the organisation and delivery of a large-scale international healthcare AI hackathon in a fast-paced environment.
 - Assisted participants by resolving queries, providing guidance, and enhancing overall participant experience.
 - Contributed to the technical evaluation of an NLP model using a Bengali dataset, reviewing the accuracy of translating medical check-up data into English and ensuring clinical relevance.
 - Contributed to event coordination, including setup, logistics, and real-time operational support.
 - Conducted interviews with student teams and industry partners, gaining insights into solution design and problem-solving approaches.
 - Supported talent matching activities between participants and industry stakeholders.
 - Captured key event moments and supported documentation of major activities.

LEADERSHIP

2025 - present	Postgraduate Student Ambassador, University of Birmingham
2025 - present	Student Rep, MSc Bioinformatics, University of Birmingham
2023 - 2025	Class Representative, Department of Genetics, University of Calcutta
2022 - 2023	Secretary of Science & Environment Club, Bethune College Students' Committee (BCSC), Bethune College

SKILLS

Programming Languages	R, Python, Bash, Awk
Bioinformatics Tools	- Proficiency in molecular docking and real-time simulation tools (e.g., UCSF Chimera, PyRx, AutoDock Vina 1.2.5, ClusPro 2.0, Discovery Studio (BIOVIA)) - Familiar with in silico ncRNA chracterization tools (e.g., RNAfold, LncRRISearch, LncBase v.3, lncLocator, iLOC-LncRNA(2.0), CPC2, RNAinter v4.0, NPInter v5.0) - Familiar with primer tools (e.g., NEBcutter V2.0, Primer3 Input) - Done working with plant regulome analysis tools (e.g., PlantPAN 4.0, Phytozome v13)
HPC Skill	Gained experience in running large-scale genomics workflows in BlueBEAR HPC (University of Birmingham)
Wet laboratory Skills	- Proficient experience in PCR, qPCR, RNA, gDNA & plasmid DNA isolation, cDNA synthesis, native and SDS PAGE, agarose gel electrophoresis, RFLP, bacterial culture, plant tissue culture, and Karyotyping - Basic experience in Agroinfiltration, Y1H assay, Y2H assay - Gained experience in handling autoclave, centrifuge, vortex, nanodrop & spectrophotometer
Digital Skills	Microsoft Office tools, Canva, Antigravity, VS Code, Google Colab, R Studio, Jupyter, Cytoscape

LANGUAGE QUALIFICATIONS

IELTS, Overall Band Score: 7.5 (Listening: 8.0, Reading: 8.0, Writing: 7.5, Speaking: 7.0), CEFR level: C1 (2025)
Goethe Start Deutsch A1, Goethe-Institut, Score: 78/100, Befriedigend (2024)

PROFESSIONAL MEMBERSHIP

IEEE (Institute of Electrical and Electronics Engineers) Member (2025 - present)
- Member of IEEE Young Professionals

COURSES & CERTIFICATIONS

Python for Genomic Data Science from Johns Hopkins University, Coursera (2025)
Introduction to Python Programming from University of Pennsylvania, Coursera (2025)
Hands-on Biotechnology Training (Jul - Dec) as a Member at Biotechnology lab in BITM Kolkata (2022)
Immunology, Indian Institute of Technology (IIT) Kharagpur, NPTEL (2021)

COMPETITIONS

LOKNITI, Policy Conclave 2021 (2021)
• Organised by Public Policy & Opinion Cell, Indian Institute of Technology (IIT) Kanpur
• **Project:** Proposed and presented a sustainable technology proposal, which is the **ACCEPTaB** (Activated, controllable, cost-effective, eco-friendly, perennial, thermo-absorbent Bio-generator)
• **Position:** 1st

ENVITIATIVE, Public Policy Hackathon 2021 (2021)
• Organised by Environment Cell, University Business School, Punjab University
• **Project:** ACCEPTaB
• **Position:** 3rd

ACADEMIC ACHIEVEMENTS

Jyotsna Manna Memorial Prize for highest proficiency in Zoology Honours in BSc Exams from Bethune college (2024)
Kamalmani Sarma Memorial Prize for highest proficiency in Zoology Honours in BSc Exams from the college (2024)
Minati Banik Memorial Prize for highest proficiency in Zoology Honours in BSc Exams from Bethune College (2024)
Special Staff Prize for securing First class in Zoology Honours in BSc Exams from Bethune College (2024)
Special College Prize for securing 1st position in Zoology BSc Honours Semester I-IV Exams in Bethune College (2024)
Chandramukhi Basu Memorial Gold Medal for being the best all-rounder student of Bethune College (2023)
Nalini Das Silver Medal for being the best all-rounder student of Bethune College (2023)

EXTRACURRICULUM

Participation in the cultural programme at the Jahresabschlussfeier (yearly social) at Goethe-Institut Kolkata (2024)
Participation in Conservation Entrepreneurship workshop, RUSA 2.0, Alipore Zoological Garden (2024)
Active Participation in Science & Environment Club 2023, BCSC, Bethune College (2023)
Active Participation in Literary Club 2021, BCSC, Bethune College (2022)
Participation in HISTORY UNVEILED, a national essay writing competition, by JIM Indore (2021)
Secured 3rd position in IMPULSE, an Instant Speech Competition 2021, by Heramba Chandra College (2021)
Participation in SIYAAHI (English), a Writing Competition, by Heramba Chandra College (2021)